

NEWTON'S INTERPOLATION FORMULA AND SUMS OF POWERS

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ABSTRACT. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the forward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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1. AUXILIARY LEMMAS

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [1], because it is simple and beautiful

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

We utilize the following notation for rising factorials

Definition 1.2 (Rising factorials). *For integers n, k*

$$n^{(k)} = \begin{cases} 0, & \text{if } k < 0, \\ 1, & \text{if } k = 0, \\ \prod_{j=0}^{k-1} (n + j), & \text{if } k > 0. \end{cases}$$

We encourage the audience to read J. F. Steffensen's work on the definition of generalized factorials [2].

Lemma 1.3 (Forward Newton's formula). *Let $f: \mathbb{Z} \rightarrow \mathbb{C}$ be a function, and let $t \in \mathbb{Z}$. Then, for all $n \in \mathbb{Z}$,*

$$f(n) = \sum_{k=0}^{\infty} \frac{(n-t)^{(k)}}{k!} \Delta^k f(t) = \sum_{k=0}^{\infty} \binom{n-t}{k} \Delta^k f(t),$$

where the k -th forward difference $\Delta^k f(t)$ of f evaluated at t is defined by

$$\Delta^k f(t) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(t+j).$$

Proof. Originally revealed by Sir Isaac Newton in [3]. Expressed in modern mathematical notation by J. F. Steffensen in [4, section 2, eq. (19)]. By using identity $\frac{(n-t)^{(k)}}{k!} = \binom{n-t}{k}$, we get RHS. $\square$

Lemma 1.4 (Forward Newton's formula for powers). *For non-negative integers n, m , and an arbitrary integer t*

$$n^m = \sum_{k=0}^m \binom{n-t}{k} \Delta^k t^m.$$

Proof. Special case of Lemma (1.3) for powers. $\square$

Lemma 1.5 (Generalized hockey-stick identity). *For arbitrary integers a, b and j ,*

$$\sum_{k=a}^b \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have, $\sum_{k=a}^b \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right)$. By hockey-stick identity $\sum_{k=0}^n \binom{k}{j} = \binom{n+1}{j+1}$ yields,

$$\sum_{k=a}^b \binom{k}{j} = \left(\sum_{k=0}^b \binom{k}{j} \right) - \left(\sum_{k=0}^{a-1} \binom{k}{j} \right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. $\square$

Lemma 1.6 (Forward hockey-stick identity). *For integer $n \geq 1$, and arbitrary integers t, k, r*

$$\sum_{j=1}^n \binom{j-t+r}{k+r} = \binom{n-t+r+1}{k+r+1} - \binom{1-t+r}{k+r+1}.$$

Proof. By setting $a = 1 - t + r$, and $b = n - t + 1$ into Lemma (1.5). $\square$

Lemma 1.7 (Multifold sums of zero powers). *For non-negative integers r, n*

$$\Sigma^r n^0 = \binom{r+n-1}{r}.$$

Proof.

- (1) Let $r = 0$, then we have $\Sigma^0 n^0 = 1$, by definition (1.1).
- (2) Let $r = 1$, then we have $\Sigma^1 n^0 = \sum_{k=1}^n 1 = \binom{n}{1}$.
- (3) Let $r = 2$, then we have $\Sigma^2 n^0 = \sum_{k=1}^n \binom{k}{1} = \binom{n+1}{2}$.
- (4) Let $r = 3$, then we have $\Sigma^3 n^0 = \sum_{k=1}^n \binom{k+1}{2} = \binom{n+2}{3}$.
- (5) Similarly, for arbitrary integer r , by hockey-stick identity $\sum_{k=1}^n \binom{k}{r} = \binom{n+1}{r+1}$, yields

$$\Sigma^r n^0 = \sum_{k=1}^n \Sigma^{r-1} k^0 = \sum_{k=1}^n \binom{k+r-2}{r-1} = \binom{r+n-1}{r}.$$

This completes the proof. □

Lemma 1.8 (Upper Binomial negation). *For integers n, k*

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$$

Proof. See [5, p. 164, eq. (5.14)]. □

Convention 1.9. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x , including zero, in Concrete Mathematics [5, p. 162], and in [6].

2. INTRODUCTION

In this manuscript, we derive formulas for sums of powers by combining forward Newton's interpolation formula for powers (1.4) with hockey-stick identity (1.6). We implement the algorithm for finding closed forms of sums of powers of integers, proposed in [7, Alg. (11.2)]. More precisely, the main idea is to determine binomial basis of Newton's formula, in terms

of variants of difference operators, for example

$$\left\{ \begin{array}{l} f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) = \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{d^k f(a)}{dx^k}. \end{array} \right.$$

We may observe that each variation of Newton's formula above has its binomial basis

$$\left\{ \begin{array}{l} \frac{(x)_n}{n!} = \frac{1}{n!} x(x-1)(x-2) \cdots (x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} = \frac{1}{n!} x(x+1)(x+2) \cdots (x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} = \frac{1}{n!} \left(n + \frac{k}{2} - 1\right) \left(n + \frac{k}{2} - 2\right) \cdots \left(n - \frac{k}{2} + 1\right) = \frac{x}{n} \binom{x+\frac{n}{2}-1}{n-1}. \end{array} \right.$$

This allows us to successfully combine Newton's formula with hockey-stick type identity, to find closed forms for sums of powers of integers. In this manuscript, we focus on closed forms of powers sums, by utilizing forward Newton's formula, and hockey-stick identity (1.6).

3. MAIN RESULTS

We start our discussion from Lemma (1.4), which yields forward Newton's interpolation formula for powers. For example,

$$\begin{aligned} n^3 &= 0 \binom{n-0}{0} + 1 \binom{n-0}{1} + 6 \binom{n-0}{2} + 6 \binom{n-0}{3}, \\ n^3 &= 1 \binom{n-1}{0} + 7 \binom{n-1}{1} + 12 \binom{n-1}{2} + 6 \binom{n-1}{3}, \\ n^3 &= 8 \binom{n-2}{0} + 19 \binom{n-2}{1} + 18 \binom{n-2}{2} + 6 \binom{n-2}{3}, \\ n^3 &= 27 \binom{n-3}{0} + 37 \binom{n-3}{1} + 24 \binom{n-3}{2} + 6 \binom{n-3}{3}. \end{aligned}$$

Thus, the transition to formula for sums of powers is quite straightforward,

$$\Sigma^1 n^m = \sum_{k=0}^m \left[\sum_{j=1}^n \binom{j-t}{k} \right] \Delta^k t^m.$$

Therefore, by setting $r = 0$ into Lemma (1.6) yields closed form of ordinary sums of powers

Proposition 3.1 (Ordinary sums of powers). *For non-negative integers n, m , and an arbitrary integer t*

$$\Sigma^1 n^m = \sum_{k=0}^m \left[\binom{n-t+1}{k+1} - \binom{1-t}{k+1} \right] \Delta^k t^m.$$

We may see that the basis $\binom{j-t+r}{k+r}$ for repeatedly applied Lemma (1.6) was indeed chosen correctly, because we derived closed form for sums of powers with ease, simply setting parameter for r . For example,

Example 3.2. *For integer $0 \leq t \leq 3$, we have*

$$\begin{aligned} \Sigma^1 n^3 &= 0 \left[\binom{n+1}{1} - \binom{1}{1} \right] + 1 \left[\binom{n+1}{2} - \binom{1}{2} \right] + 6 \left[\binom{n+1}{3} - \binom{1}{3} \right] + 6 \left[\binom{n+1}{4} - \binom{1}{4} \right], \\ \Sigma^1 n^3 &= 1 \left[\binom{n}{2} - \binom{0}{2} \right] + 7 \left[\binom{n}{3} - \binom{0}{3} \right] + 12 \left[\binom{n}{4} - \binom{0}{4} \right] + 6 \left[\binom{n}{5} - \binom{0}{5} \right], \\ \Sigma^1 n^3 &= 8 \left[\binom{n-1}{2} - \binom{-1}{2} \right] + 19 \left[\binom{n-1}{3} - \binom{-1}{3} \right] + 18 \left[\binom{n-1}{4} - \binom{-1}{4} \right] + 6 \left[\binom{n-1}{5} - \binom{-1}{5} \right], \\ \Sigma^1 n^3 &= 27 \left[\binom{n-2}{2} - \binom{-2}{2} \right] + 37 \left[\binom{n-2}{3} - \binom{-2}{3} \right] + 24 \left[\binom{n-2}{4} - \binom{-2}{4} \right] + 6 \left[\binom{n-2}{5} - \binom{-2}{5} \right]. \end{aligned}$$

From example above, we may observe binomial coefficients of negative upper index, which may look unfamiliar. However, there is no reason to be afraid of, because negative upper index binomial coefficients are completely deterministic, because $\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$. By setting $t = 0$ into Proposition (3.1) yields one more well-known identity for sums of powers, mentioned in [5, p. 190], and in [8]

Corollary 3.3. *For non-negative integers n, m*

$$\Sigma^1 n^m = \sum_{j=0}^m \binom{n+1}{j+1} \Delta^j 0^m.$$

For example,

Example 3.4. *For non-negative integer n , we have*

$$\Sigma^1 n^1 = 1 \binom{n+1}{1},$$

$$\Sigma^1 n^1 = 0 \binom{n+1}{1} + 1 \binom{n+1}{2},$$

$$\Sigma^1 n^2 = 0 \binom{n+1}{1} + 1 \binom{n+1}{2} + 2 \binom{n+1}{3},$$

$$\Sigma^1 n^3 = 0 \binom{n+1}{1} + 1 \binom{n+1}{2} + 6 \binom{n+1}{3} + 6 \binom{n+1}{4},$$

$$\Sigma^1 n^4 = 0 \binom{n+1}{1} + 1 \binom{n+1}{2} + 14 \binom{n+1}{3} + 36 \binom{n+1}{4} + 24 \binom{n+1}{5},$$

$$\Sigma^1 n^5 = 0 \binom{n+1}{1} + 1 \binom{n+1}{2} + 30 \binom{n+1}{3} + 150 \binom{n+1}{4} + 240 \binom{n+1}{5} + 120 \binom{n+1}{6}.$$

From the example above, we observe the coefficients $0, 1, 2, 0, 1, 6, 6, 0, 1, 14, 36, 24, \dots$, are defined by Stirling numbers of the second kind $a_k = k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$, which is the sequence [A131689](#) in the OEIS [9]. This leads us to the notion, that Stirling numbers of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ recover naturally from forward Newton's interpolation formula for powers. By setting $t = 1$ into Proposition (3.1), we get another well-known special case

Corollary 3.5. *For non-negative integers n, m ,*

$$\Sigma^1 n^m = \sum_{j=0}^m \binom{n}{j+1} \Delta^j 1^m,$$

mentioned, for instance, in [10]. For example,

Example 3.6. *For non-negative integer n , we have*

$$\Sigma^1 n^0 = 1 \binom{n}{1},$$

$$\Sigma^1 n^1 = 1 \binom{n}{1} + 1 \binom{n}{2},$$

$$\Sigma^1 n^2 = 1 \binom{n}{1} + 3 \binom{n}{2} + 2 \binom{n}{3},$$

$$\Sigma^1 n^3 = 1 \binom{n}{1} + 7 \binom{n}{2} + 12 \binom{n}{3} + 6 \binom{n}{4},$$

$$\Sigma^1 n^4 = 1 \binom{n}{1} + 15 \binom{n}{2} + 50 \binom{n}{3} + 60 \binom{n}{4} + 24 \binom{n}{5},$$

$$\Sigma^1 n^5 = 1 \binom{n}{1} + 31 \binom{n}{2} + 180 \binom{n}{3} + 390 \binom{n}{4} + 360 \binom{n}{5} + 120 \binom{n}{6}.$$

The coefficients $1, 1, 1, 1, 3, 2, 1, 7, 12, 6, 1, 15, \dots$ is the sequence [A028246](#) in the OEIS [9]. It is interesting to note that the work [10] provide formulas for sums of powers in terms of generalized Stirling numbers of the second kind $\left\{ \begin{smallmatrix} k \\ j \end{smallmatrix} \right\}_r$, that is

Proposition 3.7 (Cereceda (2022)). *For non-negative integers n, k*

$$\Sigma^1 n^k = \sum_{j=0}^k j! \left[\binom{n+1-r}{j+1} + (-1)^j \binom{r+j-1}{j+1} \right] \left\{ \begin{smallmatrix} k \\ j \end{smallmatrix} \right\}_r, \quad (1)$$

And

$$\Sigma^1 n^k = \sum_{j=0}^k j! \left[\binom{n+1+r}{j+1} + \binom{r+1}{j+1} \right] \left\{ \begin{smallmatrix} k \\ j \end{smallmatrix} \right\}_{-r}. \quad (2)$$

Formula (1) is a special case of Proposition (3.1) with setting $t \rightarrow r$, and upper negated binomial coefficient $\binom{1-t}{k+1}$, by using upper negation formula from Lemma (1.8). We may observe that forward finite difference of powers $\Delta^k t^m$ can be expressed in terms of generalized Stirling numbers of the second kind, that is

$$\Delta^k t^m = k! \left\{ \begin{smallmatrix} m \\ k \end{smallmatrix} \right\}_t. \quad (3)$$

This leads us to the observation, that generalized Stirling numbers of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}_t$ recover naturally from forward Newton's interpolation formula for powers, evaluated in arbitrary integer t . Now, reviewing Formula (2), we notice that it is a special case of Proposition (3.1) with setting $t \rightarrow -r$, and by using Equation (3). For example, by setting $t = 4$ into Proposition (3.1) yields rather unusual formulas for sums of powers

Example 3.8. *For non-negative integer n , we have*

$$\Sigma^1 n^0 = 1 \left(\binom{n-3}{1} + \binom{3}{1} \right),$$

$$\Sigma^1 n^1 = 4 \left(\binom{n-3}{1} + \binom{3}{1} \right) + 1 \left(\binom{n-3}{2} - \binom{4}{2} \right),$$

$$\Sigma^1 n^2 = 16 \left(\binom{n-3}{1} + \binom{3}{1} \right) + 9 \left(\binom{n-3}{2} - \binom{4}{2} \right) + 2 \left(\binom{n-2}{3} + \binom{5}{3} \right),$$

$$\Sigma^1 n^3 = 64 \left(\binom{n-3}{1} + \binom{3}{1} \right) + 61 \left(\binom{n-3}{2} - \binom{4}{2} \right) + 30 \left(\binom{n-3}{3} + \binom{5}{3} \right) + 6 \left(\binom{n-3}{4} - \binom{6}{4} \right).$$

The coefficients $1, 4, 1, 16, 9, \dots$ is the sequence [A391633](#) in the OEIS [9]. As we have all required identities in our disposal, consider the case of double sums of powers. Surely, we derive its formula with ease, and gracefully, by setting $r = 1$ into hockey-stick identity from Lemma (1.6), with additional correction terms $\Sigma^r n^0$. Therefore,

Proposition 3.9 (Double sums of powers). *For non-negative integers n, m , and an arbitrary integer t*

$$\Sigma^2 n^m = \sum_{k=0}^m \left[\binom{n-t+2}{k+2} - \binom{1-t}{k+1} \Sigma^1 n^0 - \binom{2-t}{k+2} \Sigma^0 n^0 \right] \Delta^k t^m.$$

Note that $\Sigma^1 n^0 = n$, and $\Sigma^0 n^0 = 1$. By applying upper binomial negation formula from Lemma (1.8) yields

Proposition 3.10 (Double sums of powers negated). *For non-negative integers n, m , and an arbitrary integer t*

$$\Sigma^2 n^m = \sum_{k=0}^m \left[\binom{n-t+2}{k+2} + (-1)^k \binom{k+t-1}{k+1} n + (-1)^{k+1} \binom{k+t-1}{k+2} \right] \Delta^k t^m.$$

For example, by setting $t = 5$ into Proposition (3.10) yields

Example 3.11. *For non-negative integer n , we have*

$$\begin{aligned} \Sigma^2 n^0 &= 1 \left(\binom{n-3}{2} + \binom{4}{1} n - \binom{4}{2} \right), \\ \Sigma^2 n^1 &= 5 \left(\binom{n-3}{2} + \binom{4}{1} n - \binom{4}{2} \right) + 1 \left(\binom{n-3}{3} - \binom{5}{2} n + \binom{5}{3} \right), \\ \Sigma^2 n^2 &= 25 \left(\binom{n-3}{2} + \binom{4}{1} n - \binom{4}{2} \right) + 11 \left(\binom{n-3}{3} - \binom{5}{2} n + \binom{5}{3} \right) + 2 \left(\binom{n-3}{4} + \binom{6}{3} n - \binom{6}{4} \right), \\ \Sigma^2 n^3 &= 125 \left(\binom{n-3}{2} + \binom{4}{1} n - \binom{4}{2} \right) + 91 \left(\binom{n-3}{3} - \binom{5}{2} n + \binom{5}{3} \right) + 36 \left(\binom{n-3}{4} + \binom{6}{3} n - \binom{6}{4} \right) \\ &\quad + 6 \left(\binom{n-3}{5} - \binom{7}{3} n + \binom{7}{4} \right). \end{aligned}$$

The coefficients $1, 5, 1, 25, 11, 2, \dots$ is the sequence [A391635](#) in the OEIS [9]. Therefore, generalized formula for r -fold sums of powers is straightforward, repeating the same procedure as we have done for $\Sigma^1 n^m$, and $\Sigma^2 n^m$

Theorem 3.12 (Multifold sums of powers). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \Delta^k t^m.$$

Which is quite remarkable in its pure binomial form, by using Lemma (1.7)

Proposition 3.13 (Multifold binomial sums of powers). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k t^m.$$

Explicitly,

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \binom{r-t}{k+r} \binom{n-1}{0} - \binom{r-1-t}{k+r-1} \binom{n}{1} - \dots - \binom{2-t}{k+2} \binom{r+n-3}{r-2} - \binom{1-t}{k+1} \binom{r+n-2}{r-1} \right] \Delta^k t^m.$$

For example,

Example 3.14. *For non-negative integers r, n, m we have*

$$\begin{aligned} \Sigma^r n^m &= \sum_{k=0}^m \left[\binom{n+r-0}{k+r} - \sum_{s=0}^{r-1} \binom{r-s}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 0^m, \\ \Sigma^r n^m &= \sum_{k=0}^m \left[\binom{n+r-1}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-1}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 1^m, \\ \Sigma^r n^m &= \sum_{k=0}^m \left[\binom{n+r-2}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-2}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 2^m, \\ \Sigma^r n^m &= \sum_{k=0}^m \left[\binom{n+r-3}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-3}{k+r-s} \binom{s+n-1}{s} \right] \Delta^k 3^m. \end{aligned}$$

These formulas are polynomials in n , with integer coefficients which are forward finite differences of powers, evaluated at arbitrary integer t . For example, for integer $0 \leq t \leq 4$, we have

Example 3.15. For $r = 1$ and $m = 3$, Proposition 3.13 yields

$$t = 0 : \quad \Sigma^1 n^3 = n \cdot 0 + \frac{1}{2}(n + n^2) \cdot 1 + \frac{1}{6}(-n + n^3) \cdot 6 + \frac{1}{24}(2n - n^2 - 2n^3 + n^4) \cdot 6,$$

$$t = 1 : \quad \Sigma^1 n^3 = n \cdot 1 + \frac{1}{2}(-n + n^2) \cdot 7 + \frac{1}{6}(2n - 3n^2 + n^3) \cdot 12 \\ + \frac{1}{24}(-6n + 11n^2 - 6n^3 + n^4) \cdot 6,$$

$$t = 2 : \quad \Sigma^1 n^3 = n \cdot 8 + \frac{1}{2}(-3n + n^2) \cdot 19 + \frac{1}{6}(11n - 6n^2 + n^3) \cdot 18 \\ + \frac{1}{24}(-50n + 35n^2 - 10n^3 + n^4) \cdot 6,$$

$$t = 3 : \quad \Sigma^1 n^3 = n \cdot 27 + \frac{1}{2}(-5n + n^2) \cdot 37 + \frac{1}{6}(26n - 9n^2 + n^3) \cdot 24 \\ + \frac{1}{24}(-154n + 71n^2 - 14n^3 + n^4) \cdot 6,$$

$$t = 4 : \quad \Sigma^1 n^3 = n \cdot 64 + \frac{1}{2}(-7n + n^2) \cdot 61 + \frac{1}{6}(47n - 12n^2 + n^3) \cdot 30 \\ + \frac{1}{24}(-342n + 119n^2 - 18n^3 + n^4) \cdot 6.$$

Where the coefficients

- For $t = 0$ are 0, 1, 6, 6 belong to the sequence [A131689](#) in the OEIS.
- For $t = 1$ are 1, 7, 12, 6 belong to the sequence [A028246](#) in the OEIS.
- For $t = 2$ are 8, 19, 18, 6 belong to the sequence [A038719](#) in the OEIS.
- For $t = 3$ are 27, 37, 24, 6 belong to the sequence [A391552](#) in the OEIS.
- For $t = 4$ are 64, 61, 30, 6 belong to the sequence [A391633](#) in the OEIS.

4. STIRLING FORM

We have

Lemma 4.1 (Stirling finite differences). For non-negative integers k, m , and an arbitrary integer t ,

$$\Delta^k t^m = \sum_{j=0}^m \left\{ \begin{matrix} m \\ k+j \end{matrix} \right\} \binom{t}{j} (k+j)!$$

Proof. By well-known identity $t^m = \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{t}{j} j!$ □

Thus, from Theorem (3.12) follows

Proposition 4.2 (Multifold Stirling sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \sum_{j=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \left\{ \begin{matrix} m \\ k+j \end{matrix} \right\} \binom{t}{j} (k+j)!$$

By Lemma (1.7), the pure binomial form follows

Proposition 4.3 (Multifold Stirling-Binomial sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \sum_{j=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \left\{ \begin{matrix} m \\ k+j \end{matrix} \right\} \binom{t}{j} (k+j)!$$

5. EULERIAN FORM

We have

Lemma 5.1 (Worpitzky identity). *For non-negative integers t, m*

$$t^m = \sum_{k=0}^m \binom{t+k}{m} \left\langle \begin{matrix} m \\ k \end{matrix} \right\rangle,$$

where $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$ are Eulerian numbers.

Proof. See the (1883) work of J. Worpitzky [11]. □

Thus, we obtain formula for finite differences of powers, in terms of Eulerian numbers $\left\langle \begin{matrix} m \\ s \end{matrix} \right\rangle$

Lemma 5.2 (Eulerian finite differences). *For non-negative integers t, k , and an arbitrary integer t ,*

$$\Delta^k t^m = \sum_{j=0}^m \binom{t+j}{m-k} \left\langle \begin{matrix} m \\ j \end{matrix} \right\rangle.$$

Proof. By binomial recurrence $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$, and by Worpitzky identity (5.1). □

Thus, from Theorem (3.12) follows

Proposition 5.3 (Multifold Eulerian sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \sum_{j=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \binom{t+j}{m-k} \langle m \rangle_j.$$

By Lemma (1.7), the pure binomial form follows

Proposition 5.4 (Multifold Eulerian-Binomial sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \sum_{j=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \binom{t+j}{m-k} \langle m \rangle_j.$$

6. BACKWARD DIFFERENCE FORM

We have

Lemma 6.1 (Forward-Backward relation). *For non-negative integers k, m , and an arbitrary integer t*

$$\Delta^k t^m = \nabla^k (t+k)^m,$$

where

$$\nabla^k t^m = \sum_{j=0}^k (-1)^j \binom{k}{j} (t-j)^m.$$

Thus, from Theorem (3.12) follows

Proposition 6.2 (Multifold Backward sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \nabla^k (t+k)^m.$$

By Lemma (1.7), the pure binomial form follows

Proposition 6.3 (Multifold Backward-Binomial sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \nabla^k (t+k)^m.$$

7. CENTRAL DIFFERENCE FORM

We have

Lemma 7.1 (Forward-Central relation). *For non-negative integers k, m , and an arbitrary integer t*

$$\Delta^k t^m = \delta^k \left(t + \frac{k}{2} \right)^m,$$

where

$$\delta^k t^m = \sum_{j=0}^k (-1)^j \binom{k}{j} \left(t + \frac{k}{2} - j \right)^m.$$

Thus, from Theorem (3.12) follows

Proposition 7.2 (Multifold Central sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \Sigma^s n^0 \right] \delta^k \left(t + \frac{k}{2} \right)^m.$$

By Lemma (1.7), the pure binomial form follows

Proposition 7.3 (Multifold Central-Binomial sums). *For non-negative integers r, n, m , and an arbitrary integer t*

$$\Sigma^r n^m = \sum_{k=0}^m \left[\binom{n-t+r}{k+r} - \sum_{s=0}^{r-1} \binom{r-s-t}{k+r-s} \binom{s+n-1}{s} \right] \delta^k \left(t + \frac{k}{2} \right)^m.$$

8. CONCLUSIONS

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the forward Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
<code>ValidateMultifoldSumsOfPowers.txt</code>	Validates Thm. (3.12)
<code>ValidateMultifoldBinomialSumsOfPowers.txt</code>	Validates Prop. (3.13)
<code>ValidateMultifoldStirlingSums.txt</code>	Validates Prop. (4.2)
<code>ValidateMultifoldStirlingBinomialSums.txt</code>	Validates Prop. (4.3)
<code>ValidateMultifoldEulerianSums.txt</code>	Validates Prop. (5.3)
<code>ValidateMultifoldEulerianBinomialSums.txt</code>	Validates Prop. (5.4)
<code>ValidateMultifoldBackwardSums.txt</code>	Validates Prop. (6.2)
<code>ValidateMultifoldBackwardBinomialSums.txt</code>	Validates Prop. (6.3)
<code>ValidateMultifoldCentralSums.txt</code>	Validates Prop. (7.2)
<code>ValidateMultifoldCentralBinomialSums.txt</code>	Validates Prop. (7.3)

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